

Semester: January 2025 – April 2025		
Maximum Marks: 30	Examination: In-Semester Examination	Duration: 1hr.15 Min
Programme code: 03 Programme: B.Tech Electronics and Telecommunication Engineering	Class: SY	Semester: IV (SVU 2023)
Institute/School/ Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216U03C403	Name of the Course: Communication Systems	

Question No.		Max. Marks
Q1	What is significance of Noise Figure in communication system? Certain amplifier in television receiving system has a signal to noise ratio of 20 dB at input, and becomes 15 dB at output after the signal is passing through it. Calculate noise figure in ratio and decibels. <u>OR</u> What is correlated noise? Certain nonlinear amplifier is given a two input signals of 2 KHz and 12 KHz. Calculate the sum and difference frequencies which are present at the output of this amplifier for values of m and n of 2 and 3.	05
Q2	a. Draw time domain representation of AM wave (on graph paper) having modulating signal of $10 \sin(4\pi \times 250t)$ and carrier signal of $25 \cos(4\pi \times 10^6 t)$ Calculate modulation index. b. Draw the amplitude spectrum for the same signal representing all values and show calculation.	05
Q3	Draw neat labeled circuit of medium power AM modulator. What is role of tank circuit in it? Draw the output waveforms of this circuit with and without tank circuit. <u>OR</u> Draw neat labeled block diagram of phase shift method of single sideband AM generation. Mathematically prove that use of phase shifter removes one sideband from the signal.	05

Attempt ALL of the following questions:

A.What is role of RC time constant in demodulation of AM signal? Explain with diagram.

2 marks

B.Draw the neat labeled block diagram of VSB signal generation.

2 marks

C.What is role of channel communication system? Give two practical example of it.

2 marks

D.Calculate the carrier power and sideband power of AM wave in a load resistance of $100\ \Omega$ if the peak voltage of carrier wave is 100 V and modulation index is 0.4 . What is percentage of sideband power?

2 marks

E. Write meaning of each term with units in the following equation:

$$V_N = \sqrt{4RKT B}$$

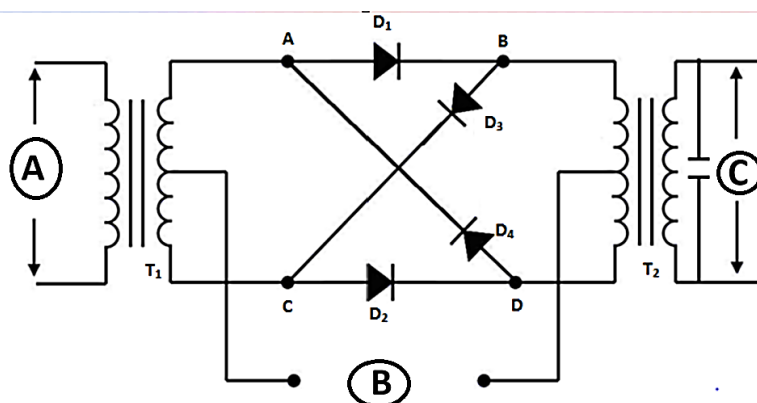
1 mark

F. Write the odd word from the list given below:

Industrial noise	Thermal noise	Shot noise	Intermodulation noise	Transit time noise
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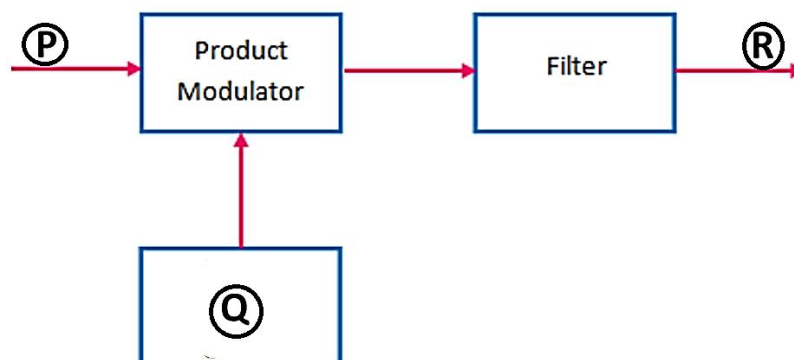
1 mark

Name the circuit given below. Draw the signals at point A, B and C giving neat labeling. Mention one advantage of using this circuit.



OR

Name the figure given below. Write the name of signals/circuits/elements present at point P, Q and R. With mathematical equations obtain the signal at point R.



Q4

Q5

05